Predicting Next-Day Stock Returns Using Linear and Nonlinear Models

Introduction and Motivation

Predicting stock price movements has always been a difficult problem in quantitative finance. In this project, we explore whether daily technical indicators, calculated from open, high, low, close, and volume data, can be used to predict the next day's return for large-cap stocks. The study focuses on five U.S. companies (AAPL, JNJ, MSFT, PG, and KO) over a four-year period from 2017 to 2020. The task is framed as a supervised regression problem.

To address this, we compare three models: ordinary least squares (OLS), support vector regression (SVR), and random forest (RF). Their performance is evaluated against a simple baseline that predicts the average return. We assess all models using three key metrics: mean squared error (MSE) to measure prediction accuracy, R-squared (R^2) to evaluate how well the models explain variation in returns, and directional accuracy (DA), which captures how often the models correctly predict the direction of price movements.

Data and Pre-processing

From the data, we created features using only information available up to day *t*, ensuring that no future information was used and avoiding look-ahead bias. The **features** include common technical indicators such as simple moving averages (SMA 10, 50, and 200), rolling volatility, relative strength index (RSI), moving average convergence divergence (MACD), Bollinger Bands, and average true range (ATR). We also included market index returns as an additional explanatory factor.

Outliers in stock and market returns can harm model performance, so they were treated using winsorization. However, this was not applied to price variables such as Open, High, Low, and Close. Prices are non-stationary and tend to grow over time, so large values are not true outliers but reflect real market behaviour. Clipping prices would remove important information and distort trends. In contrast, returns are more stable over time, comparable across periods, and their extreme values carry meaningful signals, making outlier handling appropriate.

Also, we scaled the data using **MinMaxScaler** to improve the quality of the input features. Importantly, scaling was performed separately within each cross-validation fold, meaning that parameters were learned only from the training data.

The **target variable** was defined as the next-day return.

For each asset, the dataset was **split** in time order into 75% training data and 25% test data. Model selection was carried out using time-series cross-validation with expanding windows. This strictly chronological approach ensured that future data did not influence model training or preprocessing, effectively **preventing data leakage.**

Model Development and Evaluation Design

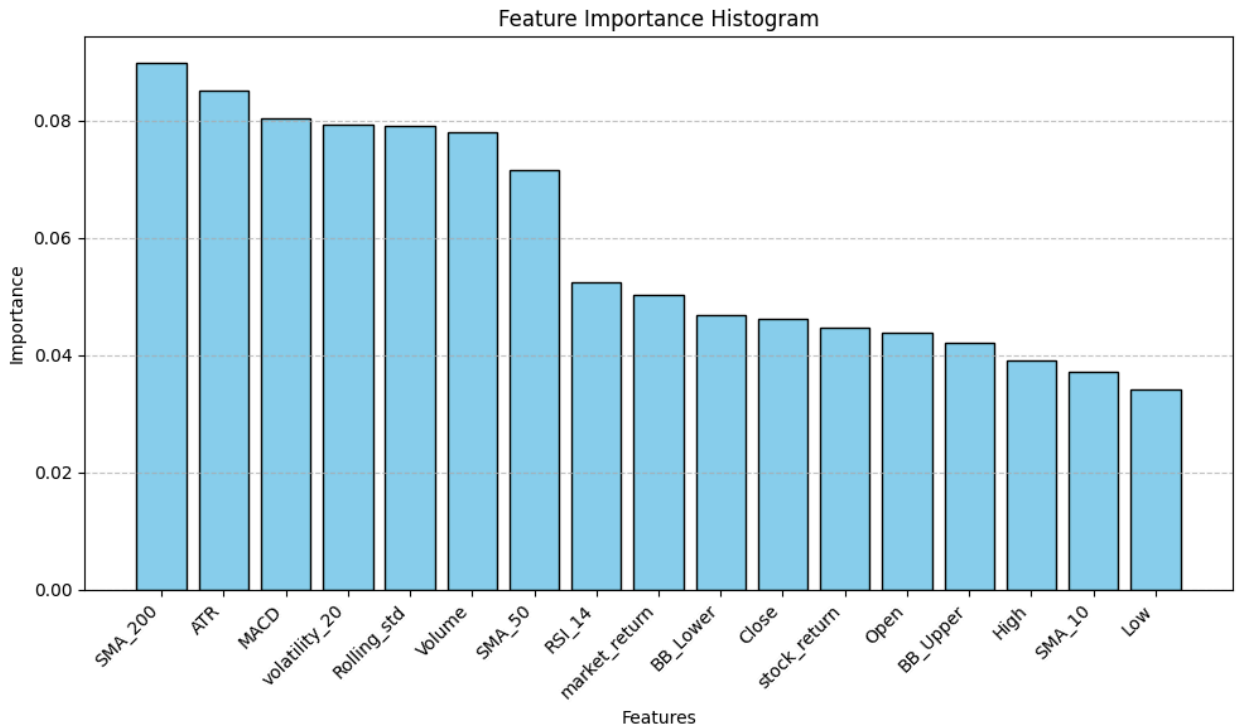
We implemented three models: (1) **OLS regression** with an intercept and all engineered features; (2) **SVR** with radial-basis-function and linear kernels, tuned via grid search on the cost and epsilon parameters; and (3) **Random Forest** regression with varying numbers of trees, maximum depths and feature fractions. A **mean predictor** served as the baseline: it always predicted the average return from the training data. Evaluation employed MSE, R^2 and DA.

Cross-validation metrics guided **hyperparameter tuning**, and the best model from each search was retrained on the full training data before testing on the hold-out set. To prevent overfitting, we compared training MSE and validation MSE with cross-validation; a large increase suggests the model has memorised noise.

Feature importance and coefficient significance

To keep the analysis concise, detailed hyperparameter tuning and interpretability steps are illustrated using AAPL as a representative example. The initial random forest model showed clear overfitting, with a very low training MSE (0.00007) compared to a much higher validation MSE (0.00056). To address this, we applied a combination of feature selection and stricter hyperparameter tuning using grid search. The final model, consisting of a small number of shallow trees, increased the training error but reduced the validation error, leading to a more balanced performance and improved generalization.

Feature importance from the tuned model identified SMA_200, ATR, volatility_20, MACD, and rolling standard deviation as the most relevant features, which were then used in the final models. An OLS model fitted on these features achieved very low explanatory power.



The OLS results support this finding. The significance of each coefficient is evaluated using p-values, which test whether a feature has a real effect on the target or if its impact could be zero. Based on this, higher volatility (captured by ATR) is associated with slightly lower next-day returns, while greater price variation shows a small positive relationship with returns. However, these effects are very weak, as the model explains only a tiny portion of the variation in returns. Similar patterns were observed across the other assets, but are not discussed here for brevity.

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                            OLS Regression Results
==============================================================================
Dep. Variable:                 target   R-squared:                       0.011
Model:                            OLS   Adj. R-squared:                  0.003
Method:                 Least Squares   F-statistic:                     1.306
Date:                Fri, 27 Mar 2026   Prob (F-statistic):              0.259
Time:                        13:34:36   Log-Likelihood:                 1529.6
No. Observations:                 604   AIC:                            -3047.
Df Residuals:                     598   BIC:                            -3021.
Df Model:                           5
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const       4.742e-05      0.004      0.011      0.992      -0.009       0.009
x1             0.0029      0.006      0.487      0.626      -0.009       0.014
x2            -0.0300      0.015     -1.993      0.047      -0.060      -0.000
x3             0.0008      0.006      0.119      0.905      -0.012       0.013
x4             0.0034      0.009      0.377      0.706      -0.014       0.021
x5             0.0190      0.009      2.142      0.033       0.002       0.037
==============================================================================
Omnibus:                       94.474   Durbin-Watson:                   2.127
Prob(Omnibus):                  0.000   Jarque-Bera (JB):             1111.912
...
==============================================================================
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Results and Discussion

Cross-asset performance

Across the five stocks, both the support vector regressor and random forest achieved slightly lower cross-validation MSE than the linear OLS model. However, all models had negative R^2 values during cross-validation, showing that none of them captured meaningful variation beyond the average return. The average MSEs were close, around 0.00030 for OLS and 0.00026 for both SVR and RF, while directional accuracy remained near 0.49, indicating almost random performance.

When evaluated on the test set, the results changed noticeably. The simple mean baseline and SVR achieved the lowest average MSE (around 0.00049), while random forest performed worse, and OLS remained the weakest. All R^2 values were still negative. In terms of directional accuracy, the baseline performed best (around 0.54).

Overall, the mean baseline remained the most reliable approach, but SVR showed competitive performance and was the strongest among the tested models.

Two key insights stand out. First, all models perform noticeably worse on unseen test data compared to cross-validation, indicating that overfitting remains an issue despite using time-aware validation methods. Second, although more complex models capture nonlinear patterns and appear stronger during training, only the SVR maintains stable performance on new data, while the random forest fails to generalize effectively.

Overall, the results suggest that simple patterns in the data are difficult to outperform, and models based only on technical indicators do not consistently improve upon basic benchmark methods.

Overfitting and Underfitting

Overfitting occurs when a model performs much better on training data than on validation or test data. This was clearly seen in the initial random forest model for AAPL, where the gap between training and validation error was very large. Although tuning reduced this gap, the model still failed to outperform the baseline on unseen data. Across all assets, a consistent pattern emerged where models achieved much lower errors during cross-validation than on the test set, confirming that overfitting remained a general issue.

In contrast, OLS showed a smaller gap between training and validation performance, but this came at the cost of very weak predictive power, indicating underfitting due to its simple linear structure. While SVR and random forest were able to capture more complex patterns, these did not translate into better performance on new data. This suggests that daily stock returns contain a high level of noise and a limited, predictable signal, making it difficult for even more flexible models to generalise effectively.

Improvements and Future Work

The results indicate that simple mean predictions were difficult to beat with the chosen models and features. Several avenues could improve performance. **Regularisation methods** (e.g., ridge or lasso regression) could reduce overfitting in linear models by penalizing large coefficients.
More informative features, such as macroeconomic indicators or news sentiment, might capture additional structure beyond price-based technicals.
In addition, **more advanced models,** such as gradient boosting, may provide a better balance between bias and variance compared to standard random forests.

Conclusion

Predicting daily stock returns from past price data remains a challenging task. In this study, we developed a set of technical indicators and applied OLS, SVR, and random forest models to five large-cap stocks. Despite careful design, including feature engineering, cross-validation, and hyperparameter tuning, the models were only able to capture very weak predictive signals. Their performance was generally close to that of a simple mean baseline, which remained the most reliable approach, although SVR showed comparable results and slight improvements for some assets.

Overall, the findings suggest that daily returns are highly noisy, and technical indicators alone provide limited useful information. For AAPL, long-term trend and volatility features appeared most relevant, but their impact was small. This highlights the need for more informative features, such as macroeconomic data, as well as improved modelling approaches.